

BANK LOAN DEFAULT RISK ANALYSIS

Data Cleaning and Feature Engineering

Complete Step-by-Step Documentation

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This document provides a complete step-by-step record of every data cleaning and feature engineering action performed on the Bank Loan Default Risk raw dataset. Raw data contained 5,100 records across 33 columns. After cleaning, the final analytical dataset contains 5,000 records and 40 columns, including 7 newly engineered analytical features.

5,100	5,000	100	33	40	7
Raw Records	Cleaned Records	Duplicates Removed	Raw Columns	Final Columns	New Features

Step 1 — Duplicate Record Removal

Action	Column Affected	Detail	Business Reason
Identify duplicates	Customer_ID	Scanned all 5,100 rows for repeated Customer_ID values. Found 100 exact duplicate records.	Each customer must appear exactly once. Duplicates cause inflated KPIs and double-counting in dashboard metrics.
Remove duplicates	Customer_ID	Removed all 100 duplicate rows. Dataset reduced from 5,100 to 5,000 unique records.	Ensures one-record-per-customer integrity across all analyses and dashboard calculations.

Step 2 — Missing Value Treatment

Column	Missing Count	Treatment Method	Replacement Value	Business Reason
Education	1,095	Categorical imputation	'Unknown'	Retains all records without making incorrect assumptions about missing qualifications.

Column	Missing Count	Treatment Method	Replacement Value	Business Reason
Employment_Status	1,016	Categorical imputation	'Unknown'	Preserves dataset size. 'Unknown' is a valid, reportable employment category in dashboards.
Occupation	150	Categorical imputation	'Unknown'	Avoids record loss and clearly highlights where occupational data is unavailable.
Annual_Income	150	Median imputation	Median of column	Median is resistant to outliers and preserves the natural income distribution better than mean.
Credit_Score	150	Median imputation	Median of column	Preserves credit score distribution without skewing risk band classifications.

Step 3 — Invalid Value Correction (Negative Financial Values)

Action	Column Affected	Records Identified	Treatment Applied	Business Reason
Fix negative income	Annual_Income	23 records	Negative values replaced with the median value of the Annual_Income column.	Income cannot be negative. Median replacement preserves the income distribution without introducing bias.
Fix negative expenses	Monthly_Expenses	25 records	Negative values replaced with the median value of the Monthly_Expenses column.	Monthly expenses must be positive. Median replacement avoids distorting disposable income analysis.
Fix negative EMI	Existing_EMI	46 records	Negative values replaced with the median value of the Existing_EMI column.	Loan repayment amounts cannot be negative. Median replacement ensures accurate debt burden calculations.

Step 4 — Text Standardisation and Category Consolidation

Column	Raw Values Found	Actions Taken	Final Standard Values
Gender	male, Male, MALE, female, Female, FEMALE	Trimmed leading and trailing spaces. Standardised capitalisation.	Male, Female

Column	Raw Values Found	Actions Taken	Final Standard Values
Marital_Status	Married, married, Single, SINGLE, Divorced	Removed extra spaces. Standardised capitalisation.	Married, Single, Divorced
Employment_Status	Employed, employed, Self Employed, Unemployed	Standardised all category labels. Consolidated variant spellings.	Employed, Self Employed, UnEmployed, Unknown
Occupation	Self Employed, Self-Employed, self employed	Consolidated all three variants into one standard label.	Self Employed
Risk_Category	High, HIGH, high, High (with extra spaces)	Trimmed extra spaces. Standardised capitalisation.	High, Low

Step 5 — Validation Checks

Action	Column Affected	Detail	Business Reason
Validate Customer_ID	Customer_ID	Confirmed all values are unique after duplicate removal. No null values remaining.	Primary key integrity is mandatory for all joins and lookups.
Validate Loan_ID	Loan_ID	Verified each loan record has a unique Loan_ID with no null values.	Loan-level tracking requires unique identifiers across all records.
Validate Age	Age	Confirmed all age values fall within a realistic range of 18 to 75. No anomalies detected.	Ages outside this range indicate data entry errors.
Validate Credit Score	Credit_Score	Confirmed all values are within the standard range of 300 to 900 post-imputation.	Credit scores outside this range are not valid and would distort risk band classifications.
Validate Loan Amount	Loan_Amount	Confirmed all loan amounts are positive. No zero or negative values remain in the dataset.	Zero or negative loan amounts are not valid financial records.
Validate Interest Rate	Interest_Rate	Confirmed all interest rates are positive and fall within the expected lending range.	Negative or zero interest rates indicate data entry errors.
Validate DTI Ratio	Debt_to_Income_Ratio	Confirmed all DTI values are positive. Checked for extreme outliers across all records.	DTI is a critical risk metric. Accuracy is essential for financial analysis.

Step 6 — Feature Engineering (7 New Columns Created)

New Column	Source Column	Logic / Bands	Business Purpose
Defaulted_Flag	Defaulted	Yes = 1, No = 0	Enables numeric KPI calculations and the default rate formula used across all dashboard metrics.
Credit_Score_Band	Credit_Score	Poor / Average / Good / Fair / Excellent	Converts the continuous credit score into interpretable risk quality tiers for segment-level analysis.
DTI_Band	Debt_to_Income_Ratio	0-20% / 21-40% / 41-60% / 61%+	Groups borrowers by debt burden level for financial stress analysis in the Financial Analysis dashboard.
Loan_Amount_Band	Loan_Amount	0-2 Lakh / 2-5 Lakh / 5-10 Lakh / 10-20 Lakh / 20 Lakh+	Enables loan exposure analysis and default rate comparison by loan size category.
Interest_Rate_Band	Interest_Rate	Below 8% / 8-10% / 10-12% / 12-14% / Above 14%	Supports rate-tier default comparison and loan pricing risk analysis.
EMI_Band	EMI	0-5,000 / 5,001-10,000 / 10,001-20,000 / 20,001-40,000 / Above 40,000	Classifies monthly repayment burden for affordability analysis and default pattern reporting.
Savings_Balance_Band	Savings_Balance	0-25,000 / 25,001-50,000 / 50,001-1 Lakh / 1-5 Lakh / 5 Lakh+	Groups borrowers by liquidity strength to evaluate the relationship between savings and default risk.

Data Quality Summary

Activity	Scope	Outcome
Duplicate Removal	100 duplicate rows	Dataset reduced from 5,100 to 5,000 unique records.
Missing Value Treatment	5 columns 2,561 total nulls	All null values resolved using median imputation for numerical columns and 'Unknown' label for categorical columns. No records were dropped.
Invalid Value Correction	3 columns 94 negative values	All negative financial values replaced with the median value of their respective columns.
Text Standardisation	5 columns	Consistent category labels enforced across all text columns. Fragmented and inconsistent labels eliminated.
Validation Checks	7 key columns	All values confirmed within expected business and statistical ranges. No anomalies remaining.

Activity	Scope	Outcome
Feature Engineering	7 new columns created	Dataset expanded from 33 to 40 columns. Fully analytics-ready for dashboard reporting and SQL analysis.